SEC issuer financial screening

Task 3 | Chunked Python processing and auditable data controls

Prepared 14 October 2026 | Retrospective report on the saved project package

Finding: coverage must precede financial interpretation

The saved run scanned 3,905,397 numerical facts from the SEC 2026 Q3 filing archive and selected 1,905 manufacturing issuers. All three screening ratios were calculable for 1,376 issuers (72.2%). Missing or quarantined facts remain visible rather than being replaced with zero.

Measure	Saved result
Eligible 10-K / 10-Q filings	1,946
Selected issuers	1,905
Retained candidate facts	8,899
Quarantined accession/tag groups	3
Issuers with all three ratios	1,376 of 1,905 (72.2%)
Reporting periods over 180 days old	9

This project demonstrates memory-conscious ingestion, explicit scope filters, one-filing-per-issuer selection, duplicate/value controls and coverage reporting. It produces a starting point for company-level review, not a credit rating or an investment recommendation.

The filing archive is organized by submission quarter. Its companies do not share a uniform quarter-end balance-sheet date. “Latest” means latest within this archive only, not across the issuer's complete filing history.

1. Pipeline and screening rules

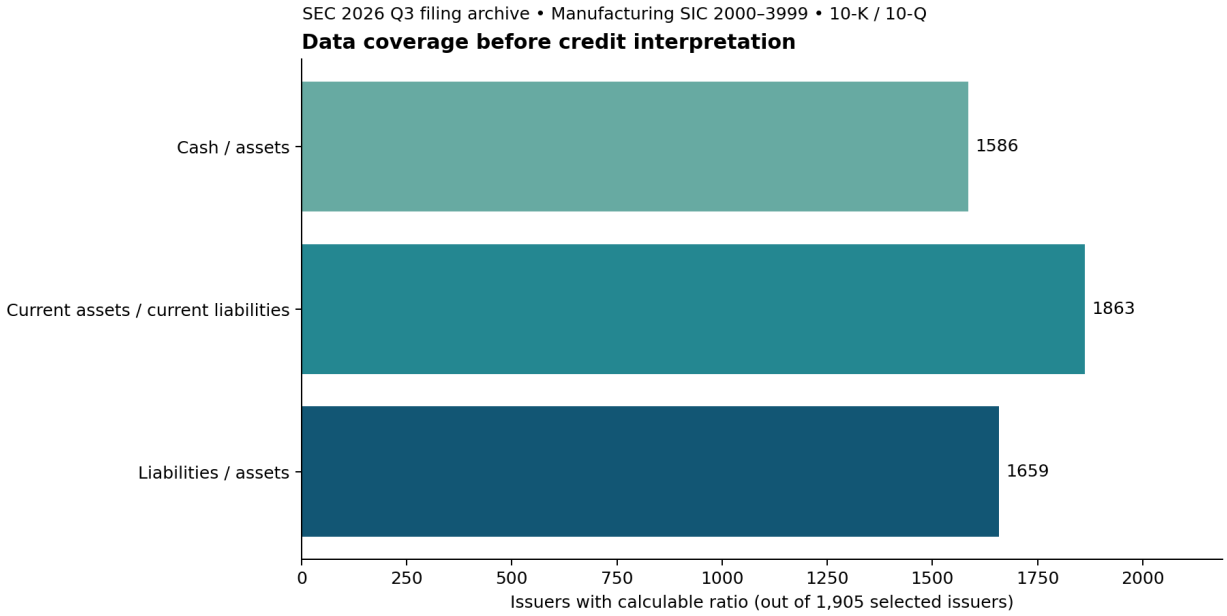
Step	Implemented rule
Select issuers	Manufacturing SIC codes 2000-3999; forms 10-K or 10-Q. Amendments are excluded.
Choose filing	Sort within CIK by reporting period, filing date, acceptance time and accession; retain the last.
Stream numerical facts	Read num.txt from the source ZIP in 100,000-row chunks, keeping only required columns.
Filter facts	Selected accession; five standard US-GAAP tags; USD; instant facts; no segments/co-registrants; matching reporting date.
Resolve values	Group by accession and tag. Any invalid or conflicting value quarantines the entire group; agreeing duplicates collapse to one fact.
Calculate ratios	Pivot clean facts, left-join all selected issuers, and calculate only with positive denominators.

The five tags are Assets, Liabilities, AssetsCurrent, LiabilitiesCurrent and CashAndCashEquivalentsAtCarryingValue. Custom taxonomy tags and alternative cash definitions are not mapped, so missingness can reflect taxonomy choices as well as unavailable data.

Ratio	Formula	Interpretation
Liabilities / assets	Liabilities / Assets	Broad balance-sheet leverage; not debt / assets.
Current ratio	AssetsCurrent / LiabilitiesCurrent	A multiple, e.g. 1.50x; not a percentage.
Cash / assets	CashAndCashEquivalentsAtCarryingValue / Assets	Cash concentration; not total liquidity.

CSV leverage and cash ratios are decimal fractions: 0.60 means 60%. A liabilities-to-assets ratio above one is not automatically a data error; it can reflect negative equity. Company context and filing review are required.

2. Coverage and row reconciliation



First exclusion or retention category	Rows
outside selected filings	3,099,738
other tag or taxonomy	782,483
non USD	45
duration fact	0
segment or coregistrant	4,694
other reporting date	9,538
retained candidates	8,899

Every scanned fact receives exactly one first exclusion reason or is retained: the categories total 3,905,397. These are sequential scope filters, not independent defect counts. Retained candidates are assessed for conflicts afterward, so quarantine counts must not be added to this reconciliation.

3. Scale, validation and reproducibility

The saved run processed 40 chunks. The uncompressed numerical file was 875,459,414 bytes (about 875 MB); the largest input chunk DataFrame was 66,346,104 bytes (about 66 MB). The recorded elapsed time was 10.76 seconds on the original execution environment. Neither the chunk measurement nor this runtime is a production capacity guarantee.

Important memory distinction: chunk DataFrame memory is not peak process RAM. The program also retains filing metadata, accumulated candidate facts, group statistics, output tables and temporary objects. Chunking reduces input pressure, but this is not a constant-memory or distributed implementation.

Recorded / coded control	Meaning
Row reconciliation	All scanned rows reconcile to first exclusions plus candidates.
Accession integrity	Duplicate SUB accession keys and orphan NUM accessions raise errors.
Issuer integrity	Final screen asserts one row per selected CIK.
Fact resolution	Invalid/nonfinite/negative or conflicting facts are quarantined; missing values stay missing.
Review flags	Nonpositive denominators, components exceeding totals and old reporting periods are exposed. No component-exceeds-total flags were recorded.

Extract BNY_Task3_Python_Pack.zip and keep data/2026q3.zip compressed. In bny_task3 run python -m pip install -r requirements.txt, then python task3.py. The --chunk-size option changes ingestion batch size. The --archive option still assumes this exercise's schema and fixed 2026 Q3 labeling; it is not a general quarterly updater.

Report evidence: README.txt, task3.py, quality_report.json, issuer_screen.csv, row_reconciliation.csv and run_log.txt. Counts and ratio coverage were checked against saved outputs for this report; no new SEC download or timed full rerun was performed.

Original source: https://www.sec.gov/files/dera/data/financial-statement-data-sets/2026q3.zip . Filing links are retained in issuer_screen.csv for manual review. Source archive SHA-256 is recorded in quality_report.json. AI-assisted coding supports implementation; the delivered files do not establish independent validation of every filing or accounting judgment.